

40.520 Stochastic Models

Spring 2026 Midterm Exam

Guideline

- **Topics:** Necessary Probability Background & Discrete Time Markov Chains
- **Date & Time:** Tuesday, March 3rd, 2:00 – 4:00 PM (Papers handed out at 1:55 pm)
- **Location:** Think Tank 22 (Building 2, Level 3, Room 2.311)
- **Permitted Materials:**
 - A Non-programmable scientific calculator
 - One double-sided A4 cheat sheet
- Scratch paper will be available upon request.
- Applications for the make-up midterm will be accepted until February 27th, 2026.

Review Points

Necessary Probability Background

- Probability axioms and basic rules
- Conditional probability and independence
- Law of Total Probability
- Bayes' Law
- Random variables: distributions, expectation (mean), and variance
- Properties of expectation and variance
- Joint distributions and marginal distributions
- Conditional expectation and the Law of Iterated Expectations
- Central Limit Theorem (CLT)
- Convergence in probability (WLLN) and almost sure convergence (SLLN) (conceptual understanding)
- Time average vs. ensemble average and the Ergodic theorem

Discrete Time Markov Chain

- Markov Property
- Define (discrete-time) Markov Chain, Transition Matrix
- n -Step Probabilities and Chapman–Kolmogorov Equations
- Random Walk and Gambler’s Ruin
- Hitting Probabilities and Hitting Times
- Stopping time
- Communication Classes and Irreducibility
- Recurrence and Transience:
 - Recurrent (Positive Recurrent vs. Null Recurrent)
 - Transient
 - Periodicity
- Stationary Distribution, Long-Run Fraction of Time, and Limiting Probabilities
- Ergodic Theorem: For an irreducible, aperiodic, positive recurrent Markov chain, the limiting probability, stationary distribution, and long-run fraction of time are equivalent.

Important Proofs

One of the following proofs covered in class will appear on the exam to encourage you to review essential material.

- For a nonnegative integer-valued random variable X , $\mathbb{E}[X] = \sum_{n=1}^{\infty} P(X \geq n)$.
- **Wald’s Equation.** Let $S = \sum_{i=1}^N X_i$, where X_i are independent and identically distributed random variables with mean $\mathbb{E}[X]$, and N is a non-negative integer-valued random variable independent of all X_i (for $i = 1, \dots, N$). Then $\mathbb{E}[S] = \mathbb{E}[X] \cdot \mathbb{E}[N]$.
- Use the Markov property to show that the future and past are conditionally independent given the present state.
- **Chapman–Kolmogorov equation.** The probability of transitioning from state i to state j in $n + m$ steps is given by $P_{ij}^{n+m} = \sum_k P_{ik}^n P_{kj}^m$.
- For the gambler’s ruin problem with an infinitely rich adversary, demonstrate that the gambler faces certain ruin when the winning probability per trial satisfies $p \leq \frac{1}{2}$.

40.520 : Stochastic Modeling Selected Problems

1. Weather and Lateness

Consider the probability that a person is late to class, which depends on the weather. The weather can be classified into three mutually exclusive and exhaustive conditions:

Rainy (R), Cloudy (C), Sunny (S)

The historical probabilities for each weather type are

$$P(R) = 0.2, \quad P(C) = 0.5, \quad P(S) = 0.3.$$

The conditional probabilities of being late (L) given the weather are

$$P(L|R) = 0.6, \quad P(L|C) = 0.3, \quad P(L|S) = 0.1.$$

Compute the probability that the person is late to class.

2. Medical Test Example

- Disease prevalence: 1 in 10,000
- Test accuracy: 95% (both true positive and true negative rates)

Compute $P(\text{Disease} \mid \text{Test positive})$

3. Expectation of a Geometric Random Variable

Let $N \sim \text{Geo}(p)$

Compute the expectation $E[N]$

4. Pittsburgh Supercomputing Center Example

Suppose job durations follow

$$X \sim \text{Exp}\left(\frac{1}{1000}\right) \text{ hours.}$$

Jobs are divided into two bins:

- Bin 1: jobs with duration < 500 hours
- Bin 2: jobs with duration ≥ 500 hours

Answer the following:

- (a) $P\{\text{Job in bin 1}\}$
- (b) $P\{X < 200 \mid \text{bin 1}\}$
- (c) Find the conditional density $f_{X|\text{bin1}}(t)$
- (d) Compute $E[X \mid \text{bin 1}]$

5. Which Exponential Happens First?

Let $X_1 \sim \text{Exp}(\lambda_1)$, $X_2 \sim \text{Exp}(\lambda_2)$ with $X_1 \perp X_2$.

Compute $P(X_1 < X_2)$.

6. Geometric Mean via Conditioning

Let $N \sim \text{Geometric}(p)$.

Compute $E[N]$ by conditioning on the outcome of the first flip.

7. Polya's Urn Model

An urn initially contains r red balls and b blue balls. At each stage, a ball is randomly selected from the urn and then returned along with m additional balls of the same color.

Let X_k denote the number of red balls drawn in the first k selections.

Compute the following:

- (a) $E[X_1]$
- (b) $E[X_2]$
- (c) $E[X_3]$

8. Central Limit Theorem

State the Central Limit Theorem.

9. Weak and Strong Laws of Large Numbers

State and prove:

- Weak Law of Large Numbers (WLLN)
- Strong Law of Large Numbers (SLLN)

10. Time Average vs Ensemble Average

Explain the difference between the **time average** and the **ensemble average**. Discuss the conditions under which the two are equal. What is an ergodic system and clearly define it.

11. Hierarchical Random Choice

Choose a number X at random from $\{1, 2, 3, 4, 5\}$. Now choose a number Y at random from the set $\{1, \dots, X\}$.

- (a) Find the joint mass function of (X, Y) .
- (b) Find the conditional mass function of X given that $Y = i$, for $i = 1, 2, 3, 4, 5$.
- (c) Are X and Y independent? Justify your answer.

12. Reinforced Urn with Random Resets

An urn initially contains one black ball and one white ball. At each discrete time step:

- A ball is drawn uniformly from the urn.
- If the ball is black:
 - with probability $1/2$ add 0 new black balls,
 - with probability $1/2$ add 1 new black ball.
- If the ball is white, remove that white ball from the urn.

After the update, with probability $1/4$, all balls are discarded and the urn resets to the initial state (one black and one white).

Let T denote the first time the urn contains no white balls.

- (a) Provide a recursive formula for $E[T]$. Define $E(b, w)$ as the expected number of steps to reach zero white balls starting with b black and w white balls.
- (b) Compute $E[T] = E(1, 1)$ by solving the resulting system of equations.
Hint: you may close the system by approximating $E(4, 1) \approx E(3, 1)$.

13. Tail-Sum Formula

Prove the tail-sum formula for integer-valued random variables:

$$E[X] = \sum_{k=1}^{\infty} P(X \geq k).$$

14. Wald's Identity

Prove Wald's Identity for a sum of a random number of i.i.d. random variables.

15. Markov Property

Prove that the past and the present are independent of each other if and only if present is conditioned.

16. Product of Independent Markov Chains

Consider two independent Markov chains $(X_n : n \geq 0)$ and $(Y_n : n \geq 0)$ defined on the state space I . Suppose (X_n) has transition matrix P and (Y_n) has transition matrix Q .

Define $Z_n = (X_n, Y_n)$.

- (a) Is $\{Z_n\}$ a Markov chain?
- (b) If so, determine its transition probabilities.

17. Chapman–Kolmogorov Equation

Prove the Chapman–Kolmogorov equation for Markov chains:

$$P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}.$$

18. Rain Markov Chain

States:

$$S_1 = \text{Rain}, \quad S_2 = \text{No Rain}$$

Transition matrix:

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{bmatrix}$$

If it is raining today, compute the probability that it will rain four days from today.

19. Two-Ball Urn Process

An urn always contains two balls (red or blue). At each step a ball is chosen randomly and replaced as follows:

- Same color with probability 0.8
- Opposite color with probability 0.2

Initially both balls are red.

Find the probability that the fifth ball selected is red.

20. Three Consecutive Heads

A fair coin is flipped independently.

Let N be the number of flips required to obtain three consecutive heads.

Compute:

$$P(N \leq 8), \quad P(N = 8).$$

21. Gambler's Ruin

A gambler starts with i dollars.

Each play:

- Win \$1 with probability p
- Lose \$1 with probability $q = 1 - p$

The gambler continues until reaching N dollars or going broke (0 dollars).

Find the probability P_i that the gambler reaches N before 0.

22. Social Mobility Markov Chain

The transition matrix describing social mobility is

$$P = \begin{bmatrix} 0.45 & 0.48 & 0.07 \\ 0.05 & 0.70 & 0.25 \\ 0.01 & 0.50 & 0.49 \end{bmatrix}.$$

Determine the long-run proportion of the population in each class.

23. Rain in Singapore

Consider the transition matrix

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}.$$

Assuming it does not rain today, compute the probability that it will rain:

- in 10 days,
- in 100 days,
- in 1000 days.

24. Ergodic Theorem

State the Ergodic Theorem for Markov chains.

25. Coin Switching Problem

Coin 1 comes up heads with probability 0.6 and coin 2 with probability 0.5. A coin is continually flipped until it comes up tails, at which time that coin is put aside and we start flipping the other coin.

- What proportion of flips use coin 1?
- If we start the process with coin 1, what is the probability that coin 2 is used on the fifth flip?

26. Flea on a Triangle

A flea moves around the vertices of a triangle in the following manner. Whenever it is at vertex i it moves to its clockwise neighbor vertex with probability p_i and to the counterclockwise neighbor with probability $q_i = 1 - p_i$, for $i = 1, 2, 3$.

- Find the proportion of time that the flea is at each of the vertices.
- How often does the flea make a counterclockwise move that is then followed by five consecutive clockwise moves?

27. Hotel Occupancy Markov Chain

Let X_n be the number of families that check into a hotel on day n , where $\{X_n\}_{n \geq 1}$ are independent Poisson(λ) random variables. Each family that checks in stays for a random number of days that follows a geometric distribution with parameter p :

$$P(\text{stay} = k) = p(1 - p)^{k-1}, \quad k \geq 1$$

All stay durations are independent of each other and of the arrivals process. Let Y_n be the number of families present in the hotel at the beginning of day n (before new arrivals on day n).

- (a) Show that $\{Y_n\}_{n \geq 1}$ is a Markov chain. Identify the state space and give an intuitive explanation.
- (b) Given $Y_n = i$, determine the distribution of Y_{n+1} , i.e. find the transition probabilities

$$P_{ij} = P(Y_{n+1} = j \mid Y_n = i).$$

28. Bonus–Malus Automobile Insurance System

In many countries in Europe and Asia, automobile insurance premiums are determined using a *Bonus–Malus* (Latin for “Good–Bad”) system. Each policyholder is assigned a positive integer-valued state, and the annual premium depends on this state. Lower-numbered states correspond to fewer claims in the past and therefore result in lower annual premiums.

Consider a hypothetical Bonus–Malus system with four states. The annual premium associated with each state is given below:

State	Annual Premium (\$)
1	200
2	250
3	400
4	600

A policyholder’s state changes from year to year depending on the number of claims made during the previous year. Based on historical data, suppose the transition probabilities between states are given by the Markov transition matrix

$$P = \begin{pmatrix} 0.6065 & 0.3033 & 0.0758 & 0.0144 \\ 0.6065 & 0 & 0.3033 & 0.0902 \\ 0 & 0.6065 & 0 & 0.3935 \\ 0 & 0 & 0.6065 & 0.3935 \end{pmatrix}.$$

- (a) Determine the stationary distribution of this Markov chain.
- (b) Using the stationary distribution, compute the average annual premium paid by a typical policyholder in the long run.

40.520 Stochastic Modeling

Feb 25, 2026

Mid-term examination

Full Marks - 55

- Attempt all questions. Questions may appear on either side of a sheet.
- The exam is 120 minutes in duration. The maximum score is 55 points.
- The use of a non-programmable scientific calculator and an A4 sheet with handwritten information on one side is allowed.
- Write your answers in this booklet. Request for extra paper only when needed.
- Return the extra sheets also when returning the exam
- Sign the SUTD honour code in the back of this page.

Question 1. ($5 \times 2 = 10$ points) Let $(X_n : n \geq 0)$ be an irreducible, discrete time Markov chain with countably infinite state space I , probability transition matrix $P = ((p_{ij})), i, j \in I$. Considering this Markov Chain, circle around the right answer (TRUE or FALSE) for questions below. Ambiguous answers and unclear markings will result in zero points. You do NOT have to justify your answer for the objective questions in Parts (a) - (e) of Question Q1.

- (a) It is possible for all states to be transient. **TRUE/ FALSE**
(b) If the equation $\pi = \pi P$ has no solution then all states are transient. **TRUE/ FALSE**
(c) If $p_{ii} = 0$ for all $i \in S$, then the chain must necessarily be periodic. **TRUE/ FALSE**
(d) If $p_{ii} > 0$ for all $i \in S$, then a stationary distribution must exist for P . **TRUE/ FALSE**
(e) An invariant distribution exists when the state space is $I := \{0, 1, 2, \dots\}$ and transition probabilities are given by,

$$p_{i,j} = \begin{cases} \frac{1}{2} & \text{if } i \geq 2 \text{ and } j \in \{i-1, i+1\}, \\ 1 & \text{if } i+j = 1. \end{cases}$$

TRUE/ FALSE

Question 2. ($7 + 7 + 1 = 15$ points) Smith is in jail and has 3 dollars; he can get out on bail if he has 8 dollars. A guard agrees to make a series of bets with him. If Smith bets A dollars, he wins A dollars with probability 0.4 and loses A dollars with probability 0.6. Find the probability that he wins 8 dollars before losing all of his money if,

- a) he bets 1 dollar each time (timid strategy);

- b) he bets, each time, as much as possible but not more than necessary to bring his fortune up to 8 dollars (bold strategy);
- c) which strategy gives Smith the better chance of getting out of jail?

Question 3. (3 + 5 = 8 points) Suppose the vertices of a triangle are labelled 1, 2 and 3, and Tom (a cat) and Jerry (a mouse) move around these vertices in the following way. If Tom is at vertex i and Jerry is at vertex j , where $i \neq j$, then Tom will move to vertex j with probability $p \in (0, 1)$ and to the other vertex, say k , with probability $1 - p$. If Jerry sees Tom moving to vertex j , then Jerry instantaneously jumps to vertex k ; otherwise, Jerry remains at vertex j (so Tom never catches Jerry).

- a) Let X_n denote the location of Jerry after n movements of Tom. Does $(X_n : n \geq 0)$ form a Markov chain?
- b) Determine the probabilities of finding Jerry in Vertices 1, 2, and 3.

Question 4 (4 + 2 = 6 points) A saleswoman flies around between Singapore, Bangkok, and Jakarta as follows. From Singapore she goes to Bangkok or Jakarta with probability $1/2$ each; from Bangkok she goes to Singapore with probability $3/4$ and to Jakarta with probability $1/4$; from Jakarta she always goes back to Singapore.

- a) What is the probability of finding the saleswoman in Singapore?
- b) What fraction of her trips are from Bangkok to Singapore?

Question 5. (8 points) Suppose that $0 < p < q := 1 - p < 1$ for the discrete-time Markov chain model on the state-space $\{0, 1, 2, 3, \dots\}$ with transition probabilities given by $p_{0,1} = 1$, $p_{i,i+1} = p$ and $p_{i,i-1} = q$ for $i = 1, 2, \dots$. Find the expected time it takes for the chain to reach state '0' when the chain is began at state '1' at time 0.¹

Question 6. (8 points) Let $(X_n : n \geq 0)$ be Markov(λ, P). Suppose that we can observe the chain X_n only when it moves. So the observable process $(Z_m : m \geq 0)$ is given by

$$Z_m = X_{S_m},$$

where $S_0 = 0$ and

$$S_{m+1} = \min\{n \geq S_m : X_n \neq X_{S_m}\}.$$

Is $(Z_m : m \geq 0)$ a Markov chain? Give reasons for your answer. Identify the transition probabilities if you argue it to be a Markov chain.

¹Hint: **Question 5** – Let k_i be the expected time it takes to reach state 0 when the Markov chain is initiated at state i . If you try to do first-step analysis, you will get an infinite system of linear equations. To solve the equations, you may use the symmetry in the problem to guess how k_i should behave as a function of i ? In particular, how many times you have to add k_1 to get k_2 ? How many times you have to add k_1 to get k_3 ?, and so on.